

Fitting a connectome GNN to a calcium observable: what was tried, what failed, and what works

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1. The problem and the earlier attempts (connectome-gnn-ca)

Generator. Flyvis integrates, for 13,741 neurons at $\Delta t = 20$ ms,

$$\tau_i \dot{v}_i = -(v_i - V_i^{\text{rest}}) + \sum_j W_{ij} \text{ReLU}(v_j) + I_i(t), \quad (1)$$

optionally with per-step process noise σ added to v .

Model. The GNN assumes a first-order, additively-aggregated class,

$$\hat{v}_i = f_\theta(v_i, \mathbf{a}_i, m_i, I_i), \quad m_i = \sum_j \widehat{W}_{ij} g_\phi(v_j, \mathbf{a}_j)^2, \quad (2)$$

learning f_θ, g_ϕ , embeddings $\mathbf{a}_i$ and $\widehat{\mathbf{W}}$. It is scored by $R_{\widehat{\mathbf{W}}}^2$ against the true $\mathbf{W}$ after gain correction.

Observable. The GCaMP6f indicator convolves the voltage,

$$C_i(t) = (K * v_i)(t), \quad K(t) \propto \left(1 - e^{-t/\tau_r}\right) e^{-t/\tau_d}, \quad (3)$$

with $\tau_r = 75$ ms, $\tau_d = 400$ ms, support 2.4 s, i.e. $L = 120$ samples.

What was tried. Every earlier run set `observable: calcium`, which substitutes C for v everywhere in (2): the model is asked to satisfy $\dot{C}_i = f_\theta(C_i, \mathbf{a}_i, \sum_j \widehat{W}_{ij} g_\phi(C_j)^2, I_i)$.

config	observable	σ	recurrent	outcome
<code>flyvis_noise_free_kernel</code>	calcium	0	no	$R_{\widehat{\mathbf{W}}}^2$ from -18.0 to -0.0101 (50-iter sweep)
<code>..recurrent_4</code>	calcium	0	$t_s=4$	rollout $r = -0.153$
<code>..recurrent_8</code>	calcium	0	$t_s=8$	rollout $r = -0.220$
<code>..recurrent_16</code>	calcium	0	$t_s=16$	rollout $r = -0.277$
<code>..kernel_noise_00{1,2,3}</code>	calcium	0	no	deconvolution entry point, $\gamma = 0.01/0.02/0.03$

An automated 50-iteration hyperparameter search swept `lr.W` (3×10^{-4} to 1.5×10^{-3}), `lr`, `lr_embedding`, W initialisation (`randn_scaled`, `zeros`), L_1/L_2 on W and f_θ , and recurrent rollout depth. It found exactly two regimes and no third:

- **W frozen.** With `coeff_WL1` = 0.14, $R_{\widehat{\mathbf{W}}}^2 = -0.0101$ *exactly*, reproducibly across $t_s \in \{1, 2, 4\}$ and two learning rates. That value is $\widehat{\mathbf{W}}$ never leaving its initialisation.
- **W destabilised.** With $L_1 = 0$, $\widehat{\mathbf{W}}$ moves but diverges: $R_{\widehat{\mathbf{W}}}^2 = -1.25$ (recurrent), -3.6 to -18.0 (one-step).

The search log records “no genuine W recovery after 45 iterations”. Notably f_θ *did* fit ($R^2 = 0.75 \rightarrow 0.92$ as rollout depth grew) while $\widehat{\mathbf{W}}$ stayed pinned: the loss was satisfiable without recovering the weights.

Why this could not have worked. It is a structural mismatch, not a tuning failure. Differentiating (3) gives

$$\dot{C}_i = K * \dot{v}_i, \tag{4}$$

so $\dot{C}_i(t)$ depends on the *entire history* of $\dot{v}$ over the kernel support, whereas (2) is Markovian in the observed state. The generator of C is not first-order in C , so no $(f_\theta, g_\phi, \widehat{\mathbf{W}})$ represents it. Recurrent training was the right instinct but an order of magnitude short: it supplies $t_s \leq 16$ steps against a kernel needing $L = 120$. A second, independent handicap: those runs used $\sigma = 0$, which is the degenerate regime where $\mathbf{W}$ is underdetermined even from *perfect* voltage ($R_{\widehat{\mathbf{W}}}^2 = 0.89$ vs 0.99 at $\sigma = 0.05$).

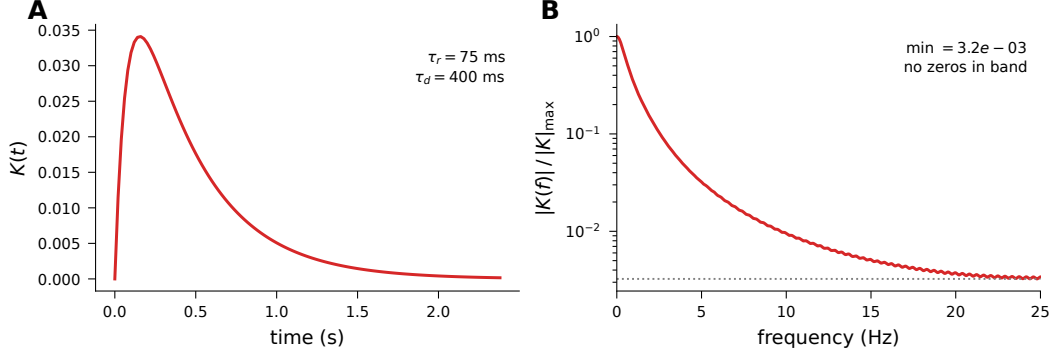


Figure 1: The indicator kernel is *given*: the inverse is built from the same config fields the generator used (τ_r , τ_d , length, Δt), so K is known exactly rather than estimated. **A** $K(t)$, 120 samples over 2.4 s, peak at 0.16 s, unit sum. **B** its transfer function. $|K(f)|$ falls by $308\times$ from DC to Nyquist — which is why the calcium trace looks smooth — but it has *no zeros in band* (minimum 3.2×10^{-3} of peak, dotted). A convolution with no spectral zeros is exactly invertible, so the high-frequency content is scaled down, not destroyed. The worst-case amplification of $308\times$ against float32 precision ($\sim 10^{-7}$) puts the noise-free error floor near 3×10^{-5} , far below the signal. That is the whole reason panel C of Fig. 2 is possible, and equally the reason it collapses once photon noise fills the attenuated band.

2. Current implementation and results

Observation model. Four degradations composed on one set of trajectories at $\sigma = 0.05$ and *zero* measurement noise unless stated: (i) kernel (3); (ii) indicator saturation, a Hill flattening past K_d set at $1.57\times$ the trace SD; (iii) a 500 ms exposure (2 Hz); (iv) shot noise matched in SNR to the paper’s γ rows.

Fitting the raw trace. With observable: calcium at $\sigma = 0.05$ the frozen- W regime reappears at every degradation. The oracle, which fits only the constants of (1), retains partial recovery.

observable	GNN $R_{\widehat{W}}^2$		Known-ODE $R_{\widehat{W}}^2$	
	best	final	best	final
voltage (control)	0.984	0.983	—	—
calcium, kernel only	−0.010	−0.010	0.373	0.332
+ saturation	−0.010	−0.010	0.077	0.006
+ 2 Hz exposure	−0.010	−0.010	0.278	0.277
+ shot noise $\gamma = 0.1$	−0.010	−0.010	−0.009	−0.379
+ shot noise $\gamma = 0.2$	−0.010	−0.010	−0.009	−0.325
all composed	−0.010	−0.010	−0.009	−0.099

“best” = maximum over training, “final” = value at 1.52M iterations. Both are given because they differ, and the difference is itself a result (§2, *decay*). Note that “best” for the GNN and for the oracle on the shot-noise rows is ≈ -0.01 , which is $\widehat{W}$ never leaving its initialisation: those arms never recovered anything at any point in training, rather than recovering and then losing it.

The fix: invert first, then fit. Deconvolve C back to a voltage estimate and train an ordinary voltage arm on it, restoring the Markov structure (2) assumes (Fig. 2). The inverse is Tikhonov-regularised in Fourier space,

$$\widehat{V}(f) = \frac{K^*(f) F(f)}{|K(f)|^2 + \lambda \max |K|^2 |R(f)|^2}, \quad R(f) = |1 - e^{-i\omega}|^2. \quad (5)$$

Deconvolution quality vs. photon noise. λ retuned per level by maximising $r(\dot{v})$, so each row is the best inversion available:

$\gamma/\text{SD}(C)$	SNR (dB)	λ	r on v	r on $\dot{v}$
0	∞	10^{-6}	0.9998	0.993
0.01	40.0	10^{-3}	0.980	0.435
0.03	30.5	10^{-2}	0.970	0.280
0.10	20.0	10^{-1}	0.956	0.172

Two-photon GCaMP sits at roughly 20–40 dB, i.e. $\gamma/\text{SD} \approx 1\text{--}10\%$. The trace stays faithful ($r \geq 0.96$ throughout) while its derivative is already more than halved at 40 dB. Since $\mathbf{W}$ is read from the derivative, a high rollout r on a calcium trace is consistent with the weights being entirely unrecoverable.

Fits on the deconvolved observable (in progress). Eight runs, both arms $\times$ four noise levels. Landed so far:

Eight runs, both arms $\times$ four noise levels. The oracle arms have finished (1.52M iterations); the GNN arms are **still training** and are reported at 880k, so their “final” column is a running value, not a converged one.

γ/SD	SNR (dB)	GNN $R_{\widehat{W}}^2$ (at 880k)		Known-ODE $R_{\widehat{W}}^2$	
		best	current	best	final
0	∞	0.967	0.961	0.915	0.489
0.01	40.0	0.144	0.144	0.405	−0.162
0.03	30.5	−0.011	−1.234	0.181	−0.159
0.10	20.0	−0.010	−0.791	−0.009	−0.142

The noise-free row is a *control*, not a result: with a known kernel and no measurement noise, (3) is exactly invertible and recovering v is division in Fourier space. A high score there confirms the pipeline; it says nothing about calcium imaging. The rows that carry the argument are 40/30/20 dB, and there recovery is gone for both arms: the best the oracle reaches at 40 dB is 0.405, and by 20 dB neither arm ever leaves initialisation.

Decay, and why “best” is not free. On the deconvolved observable the oracle does not converge — it peaks early and then drifts away from the truth, 0.915 at 32k falling monotonically to 0.489 at 1.52M. This does *not* happen on true voltage, where oracle runs rise monotonically and end at their maximum (0.977–0.979, **last** = **max** across five CV folds). The likely mechanism is that the deconvolved trace is near voltage but not identical ($r = 0.9998$; derivative amplitude 0.93), and the oracle, having only $\tau, V^{\text{rest}}, \mathbf{W}$ to fit, must absorb that residual mismatch into $\mathbf{W}$; the GNN can absorb it into f_θ, g_ϕ instead and its $R_{\widehat{W}}^2$ plateaus rather than decaying. *This mechanism is inferred from the pattern and has not yet been measured* — the direct test is whether the fitted τ and V^{rest} drift in step with the $R_{\widehat{W}}^2$ decay.

Two cautions on the “best” columns. Selecting the maximum over training is early stopping on ground truth, which is unavailable in any real application — the same objection the paper raises against its own agentic hyperparameter search. And at 30/20 dB the GNN also decays (best −0.011, current −1.234), so the decay is not unique to the oracle; it is unique to a *misspecified observable*, and it appears in whichever model lacks the capacity to absorb the mismatch at that noise level.

3. Why the first deconvolution failed, and what to fix

The dataset builder hard-coded $\lambda = 3 \times 10^{-3}$ with the *derivative* regulariser. In (5) that sets $R(f) = |1 - e^{-i\omega}|^2$, which is zero at DC and maximal at Nyquist, so the penalty falls entirely on the high band — exactly the band $\mathbf{W}$ is identified from. Since K is itself a low-pass, the result is a *double* low-pass: the indicator attenuated the high frequencies and the regulariser attenuated them again instead of restoring them. Measured cost: derivative amplitude $0.28\times$ truth and $r(\dot{v}) = 0.43$, while $r(v)$ stayed at 0.98. The fitted $R_{\widehat{W}}^2 = 0.07$ therefore characterised the preprocessing, not the observation model, and is withdrawn.

The constant came from the figure path, which *injects* synthetic noise when none is present ($0.20\times$ per-neuron SD) and then sets $\lambda = 100\sigma^2$. Heavy smoothing is correct for a noisy trace; the builder copied the constant while feeding it a clean one.

To fix / to keep fixed.

1. λ must track the noise on the trace, not be a constant. It is now tuned per level by maximising $r(\dot{v})$; $\lambda \approx 10^{-6}$ noise-free.
2. Two edge artefacts must both be handled, and they are different ends. The FFT is circular, so the end of the record wraps onto its start — reflect-padding by one kernel length moves that wrap into discarded material. Separately, frames 0..3 are kernel warm-up, where C is an incomplete convolution and v is genuinely unrecoverable; those are held at the first valid frame. Omitting the padding alone cost $r(\dot{v}) = 0.97 \rightarrow 0.24$ on the test split.
3. **Score any deconvolution change on $r(\dot{v})$, never on $r(v)$.** Both defects above were invisible in $r(v)$, which moved by less than 0.02 while the derivative lost most of its content.
4. Stimulus discontinuities need no special handling: measured over 800 abrupt transitions, the error in the following 5 frames is $1.7\times$ the median and decays to $1.1\times$ by 20 frames.

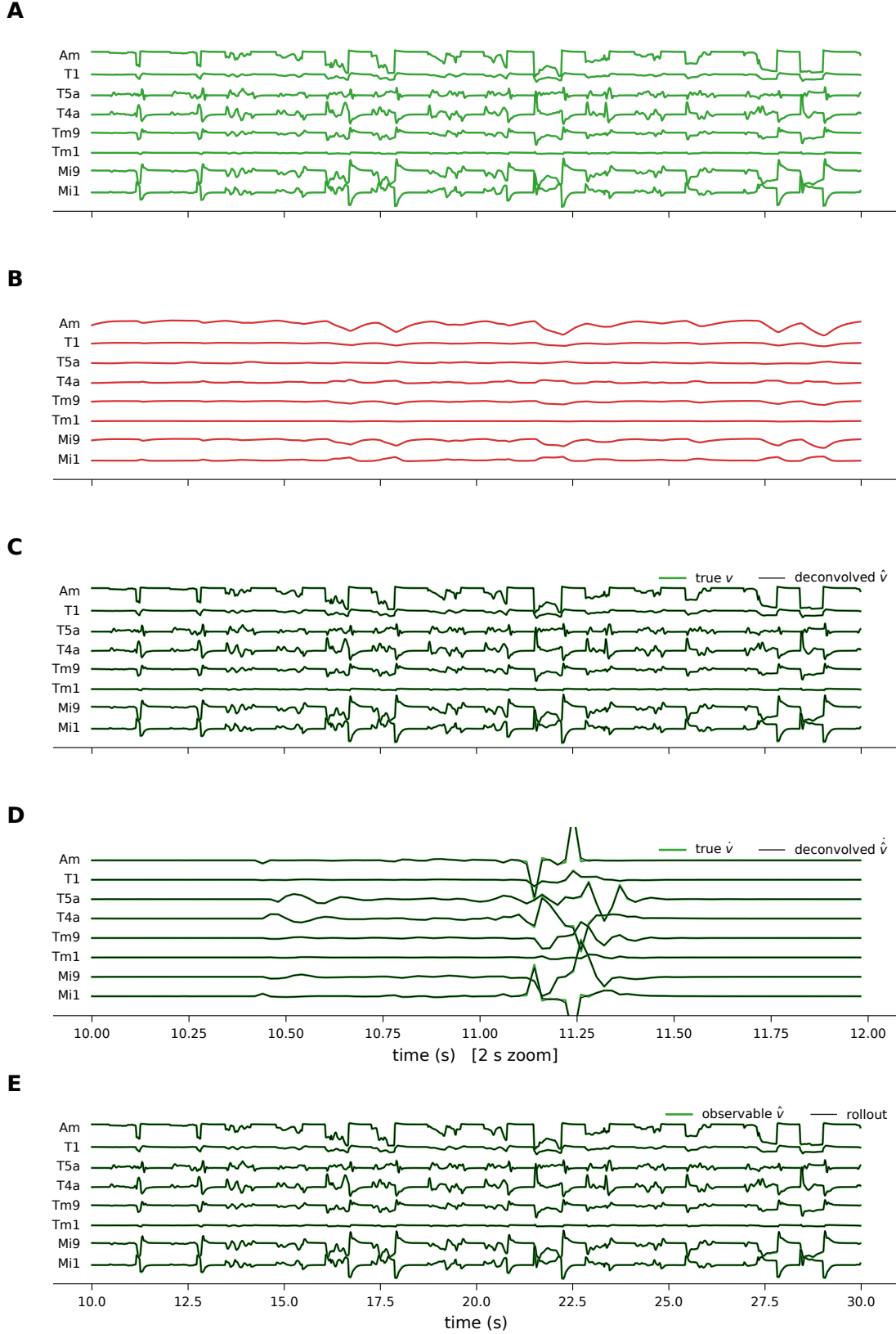


Figure 2: The observable chain, no measurement noise, eight reference cell types. All panels are the same cells over the same held-out test segment, so they can be read against each other. **A** ground-truth voltage v . **B** the GCaMP6f observable $C = K * v$, same vertical scale: the fast structure that identifies $\mathbf{W}$ is visibly gone while the slow envelope survives. Quantitatively $\text{SD}(\dot{C}) = 0.074\text{SD}(\dot{v})$, a $13\times$ attenuation — but attenuation, not deletion, which is the point, and Fig. 1B shows why. **C** the Wiener inverse (black) over the truth (green), $\lambda = 10^{-6}$: $r = 0.9999$. **D** the derivative, 2 s zoom because at $\Delta t = 20$ ms it is too dense to read over 20 s; $r(\dot{v}) = 0.996$. This is the panel that matters, since $\mathbf{W}$ is identified from $\dot{v}$ and not from v ; the deflections here at 11.15–11.25 s are the derivative of the excursions visible at the same time in A, C and E. **E** rollout of the Known-ODE arm fitted on this observable, $r = 0.9993$. Note the green trace in E is the *deconvolved observable the model was fitted on*, not the ground truth, which is the quantity a rollout is scored

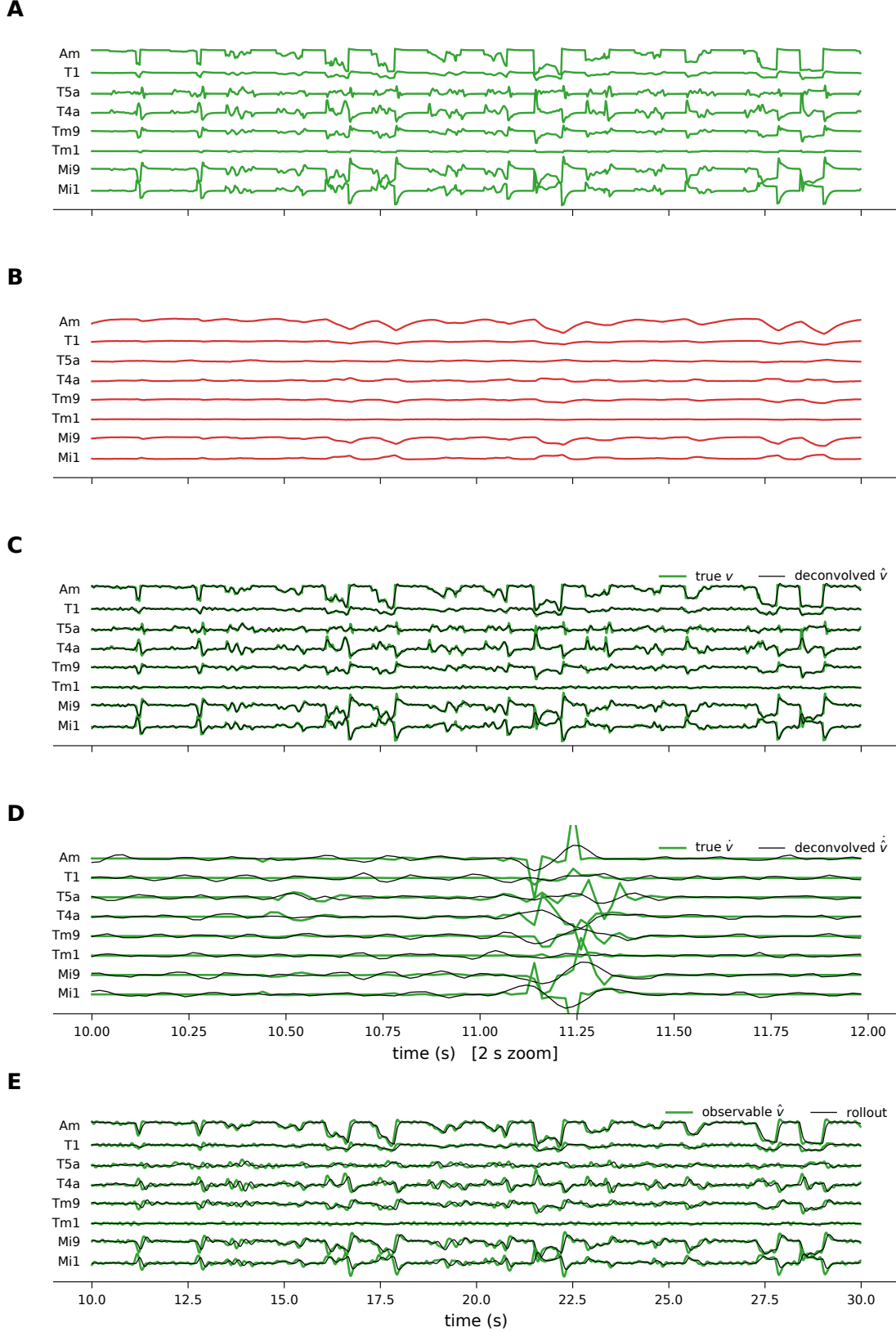


Figure 3: The same chain at $\gamma/\text{SD} = 0.01$, i.e. 40 dB — the *cleanest* end of the realistic two-photon range. Same cells, same segment, same panels as Fig. 2, so the two figures can be laid side by side. Panels A, B, C and E are essentially indistinguishable from the noise-free case: the calcium trace in B looks the same (1% noise is invisible at this scale), the deconvolved voltage in C still tracks the truth at $r = 0.987$, and the rollout in E is still $r = 0.957$. Panel D is where the difference is, and it is not subtle: the deconvolved derivative (black) departs from the truth (green) everywhere, most visibly across the transient at 11.15–11.35 s, and $r(\dot{v})$ has fallen from 0.996 to 0.545. Every panel that an experimenter would look at to judge the preprocessing still looks right; the one quantity that $\mathbf{W}$ is read from has already lost half its content. This is the figure-level statement of the whole result, and it is why $R_{\mathbf{W}}^2$ collapses at 40 dB while rollout r stays high.